

# Soft module sizing

Hard modules have fixed width and height

# The idea

- For soft module A with area  $a$
- Re-pack after each sizing move
- On our starter, A is soft; B through E stay hard
- Deadspace uses the same formula

# Browser lab track

- Open **soft-module-sizing**
- Select module A, drag aspect within bounds
- Confirm area of A stays constant while density of the packing geometry changes
- Then implement aspect moves in Track A

# Implement track

- Extend your packer so A's (w, h) can vary under aspect constraints at constant area
- Combine with a simple search, even greedy aspect tries help
- Report legality and deadspace before and after
- Reject sizes outside aspect bounds

# Pitfalls

- Using aspect limits on  $w/h$  but updating only  $w$  without fixing  $h = \text{area}/w$  breaks the area
- Allowing zero width is nonsense, enforce positive sizes
- Soft sizing cannot fix an area budget that already exceeds the outline

# Your turn

- Improve a legal packing by resizing A within bounds
- Next: hard macros that refuse to move or resize